

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**

**ASSESSMENT TOOL 2 – Traversal Challenge**

|                         |                                                                                                                                                |                      |                                         |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------|
| <b>Course Code:</b>     | CSA0309                                                                                                                                        | <b>Course Title:</b> | Data Structures                         |
| <b>CO Assessed:</b>     | CO3 – Precision (BL3, BL5)                                                                                                                     | <b>Assessment:</b>   | Assessment Tool 2 – Traversal Challenge |
| <b>Weightage:</b>       | 30% of CIA                                                                                                                                     | <b>Total Marks:</b>  | 100                                     |
| <b>CO3 Statement:</b>   | <i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.</i> |                      |                                         |
| <b>Student Name:</b>    | SRIVATSAN L                                                                                                                                    | <b>Reg. No.:</b>     | 192521480                               |
| <b>Section / Batch:</b> | 1                                                                                                                                              | <b>Date:</b>         | 31/07/2026                              |

**Code of Conduct**

*I SRIVATSAN L (192521480) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: SRIVATSAN L

**Instructions**

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

| Section / Topic | Focus                                               | Q Nos | Marks     |
|-----------------|-----------------------------------------------------|-------|-----------|
| Tree Traversals | Traversal types, In order, Post order and Pre order | 5     | 100       |
| GRAND TOTAL:    |                                                     |       | 100 Marks |

## Traversal Challenge

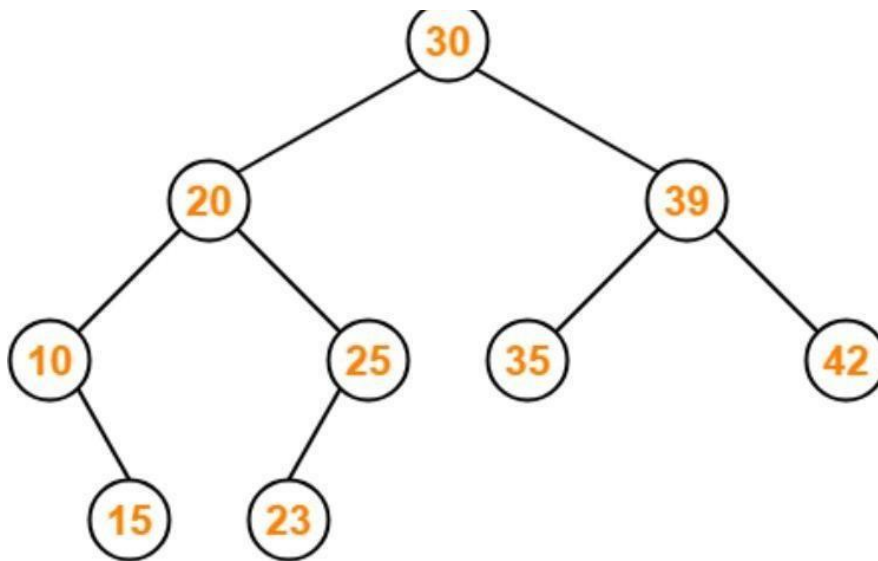
### Rubrics for Calculation

| Criteria                                  | Weightage | Level 4 – Exemplary                                                                                                 | Level 3 – Proficient                                      | Level 2 – Developing                                                     | Level 1 – Beginning                                        |
|-------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Conceptual Understanding</b>           | 20        | Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge | Shows good understanding with minor conceptual gaps       | Basic understanding; several misconceptions present                      | Limited understanding; major concepts misunderstood        |
| <b>Traversal Accuracy</b>                 | 20        | Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.                 | Completes most traversals correctly with one minor error. | Correctly performs only some traversals with multiple errors.            | Unable to perform most traversals correctly.               |
| <b>Selection of Appropriate Traversal</b> | 30        | Correctly selects the appropriate traversal for every given problem scenario.                                       | Selects the correct traversal for most scenarios.         | Selects appropriate traversal for only a few scenarios.                  | Frequently selects incorrect traversal methods.            |
| <b>Completion &amp; Time Efficiency</b>   | 20        | Completes all challenges within the allotted time while maintaining accuracy.                                       | Completes most challenges within the time limit.          | Completes only part of the challenge or exceeds the time limit slightly. | Unable to complete the challenge within the allotted time. |
| <b>Evidence Submission</b>                | 10        | Uploads complete screenshots showing successful completion, score, and student details.                             | Uploads screenshots with minor missing information.       | Uploads incomplete or unclear screenshots.                               | No valid evidence submitted.                               |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

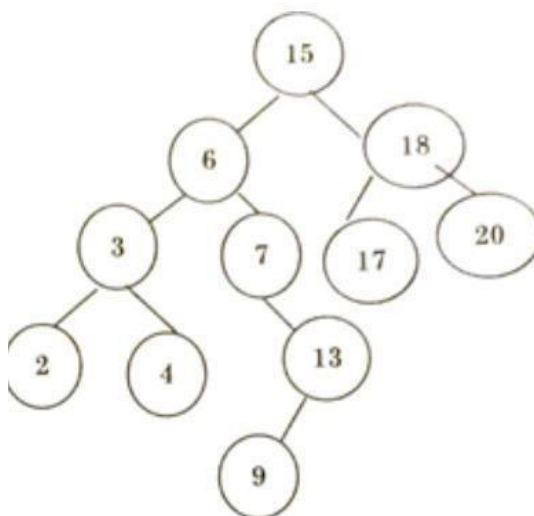
## Question No:1

Write the traversal sequences for following trees



## Question No: 2

Write the Inorder, Preorder & Postorder traversal sequences for following trees



### Question No: 3

Consider the following Inorder and Preorder traversal of a tree Inorder: D B E F A G H C

Preorder: A B D E F C G H

What is the Postorder traversal of the same?

1. D F E B H C G A

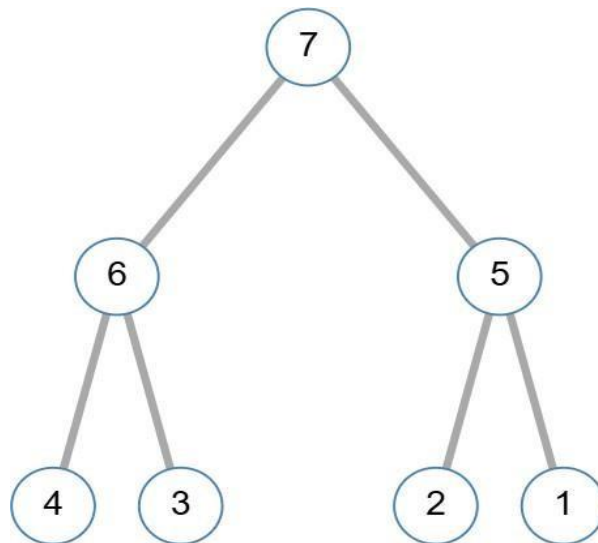
2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

### Question No: 4

**Find Inorder, Preorder and Postorder**



### Question No: 5

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

# ANSWERS

## Question No: 1

Write the traversal sequences for the following tree.

Inorder Traversal:

10, 15, 20, 23, 25, 30, 35, 39, 42

Preorder Traversal:

30, 20, 10, 15, 25, 23, 39, 35, 42

Postorder Traversal:

15, 10, 23, 25, 20, 35, 42, 39, 30

## Question No: 2

Write the Inorder, Preorder and Postorder traversal sequences for the following tree.

Inorder Traversal:

2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20

Preorder Traversal:

15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Postorder Traversal:

2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

## Question No: 3

Consider the following Inorder and Preorder traversal of a tree. Find the Postorder traversal.

Inorder: D, B, E, F, A, G, H, C

Preorder: A, B, D, E, F, C, G, H

Postorder Traversal:

D, F, E, B, G, H, C, A

**Question No: 4**

Find Inorder, Preorder and Postorder.

Inorder Traversal:

4, 6, 3, 7, 2, 5, 1

Preorder Traversal:

7, 6, 4, 3, 5, 2, 1

Postorder Traversal:

4, 3, 6, 2, 1, 5, 7

**Question No: 5**

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. Find the resulting tree's traversal sequences.

The resulting Binary Search Tree is formed by inserting the numbers according to the standard natural number ordering.

Inorder Traversal:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Preorder Traversal:

7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Postorder Traversal:

0, 2, 4, 3, 1, 6, 5, 9, 8, 7